UNIT 7 · TOPIC 2(A)

Solar Radiation and the Solar Spectrum

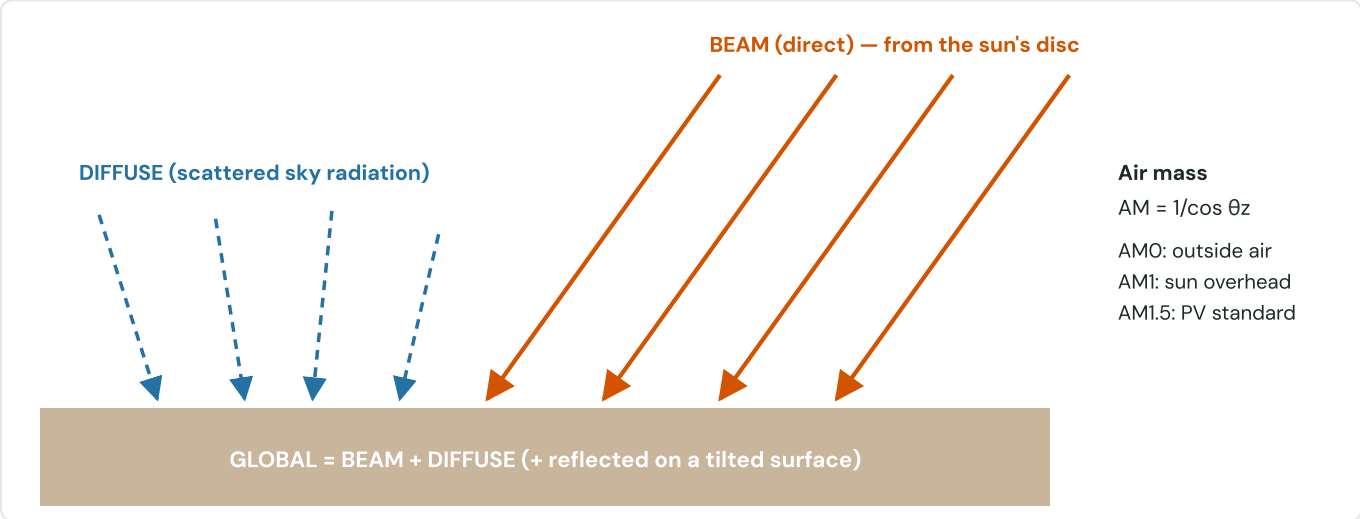
Syllabus topic: Solar radiation: measurement and characteristics of the solar spectrum

The sun and the solar constant

- The sun radiates approximately as a black body at ~**5800 K** (surface).
- **SOLAR CONSTANT (I_sc)** = solar irradiance on a surface normal to the beam, at the mean Earth–Sun distance, JUST OUTSIDE the atmosphere = **1367 W/m²** (WMO value) ≈ 1.94 cal/cm²·min = 4.92 MJ/m²·h.
- It varies about **±3.3%** over the year because of the elliptical orbit — maximum in early January, when Earth is nearest the sun.

Satellite measurements now give ~1361 W/m² (IAU 2015 nominal value); exam keys generally use 1367 W/m² (WMO) or the older 1353 W/m². The classic 1.94 cal/cm²·min corresponds to 1353 W/m²; 1367 W/m² converts to ≈ 1.96 cal/cm²·min.

Attenuation through the atmosphere



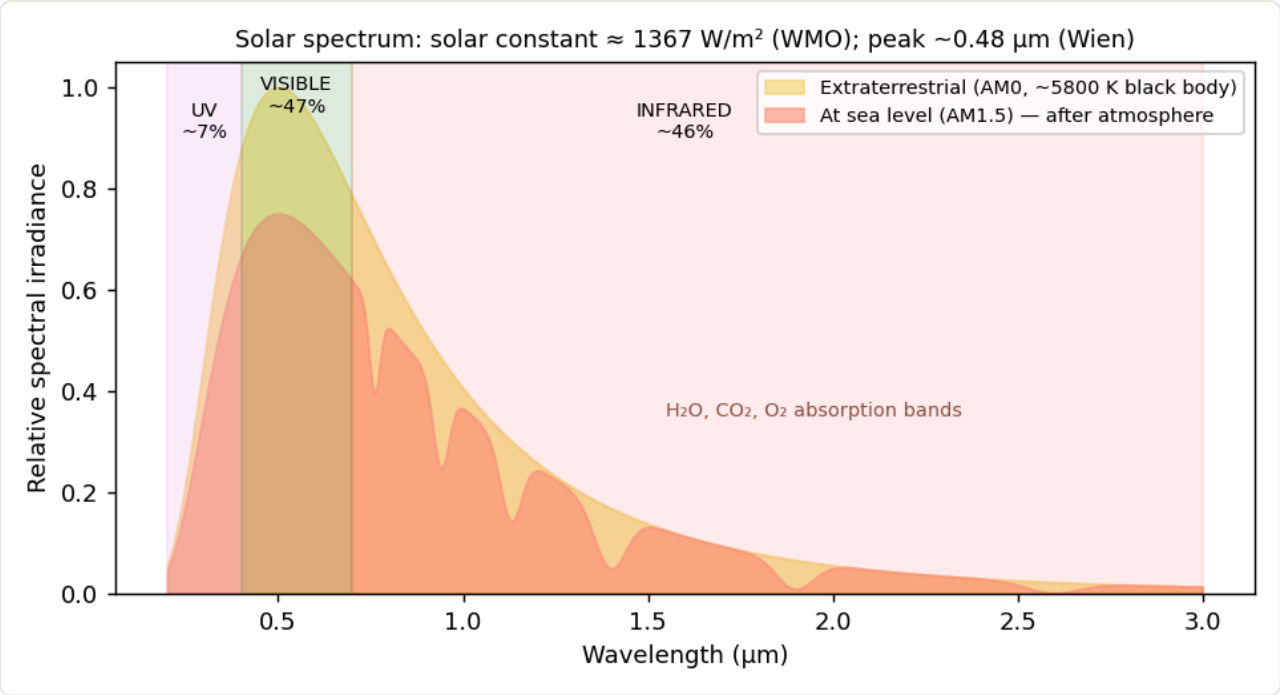
Components of terrestrial solar radiation and air mass (redrawn from the source figure)

Process	Agent
ABSORPTION	Ozone (UV); water vapour and CO_2 (infrared)
SCATTERING	Rayleigh — by molecules; Mie — by aerosols/dust; and by clouds

$$\text{Air mass } AM = 1/\cos \theta_z \ (\theta_z = \text{zenith angle})$$

- **AM0** = extraterrestrial; **AM1** = sun overhead at sea level; **AM1.5** = the standard reference for rating PV modules (~**1000 W/m²**).
- **BEAM (direct)** radiation comes straight from the sun's disc; **DIFFUSE (scattered)** radiation comes from the sky dome.
- **GLOBAL (total)** radiation = beam + diffuse (horizontal surface); on a tilted plane add ground-reflected (albedo) radiation.

Characteristics of the solar spectrum



Solar spectrum outside the atmosphere (AM0) and at sea level (AM1.5) — illustrative curves

- The extraterrestrial spectrum spans ~**0.2–3 µm** and approximates a 5800 K black body; by Wien's law its peak is near **0.48 µm** (blue-green, visible).
- At the ground the spectrum is notched by absorption bands (H_2O, CO_2, O_2), and ozone cuts the UV end.

Band	Wavelength	Share of energy (approx.)
ULTRAVIOLET	< 0.4 µm	~7%
VISIBLE (photosynthetically active, PAR)	0.4–0.7 µm	~47–48%
INFRARED	> 0.7 µm	~45–46%

PV cells respond to only part of the spectrum (silicon ~**0.4–1.1 µm**), so cell efficiency is well below the full-spectrum limit; solar-thermal collectors use the whole spectrum as heat.

Practice MCQs

Q5. The solar constant is the irradiance on a normal surface at mean Earth–Sun distance:

(a) Inside the atmosphere
(b) Just outside the atmosphere
(c) At the sun's surface
(d) At the equator only

Q6. The accepted value of the solar constant (WMO) is about:

(a) 137 W/m²
(b) 1367 W/m²
(c) 5800 W/m²
(d) 100 W/m²

Q7. In cal/cm²·min, the solar constant is approximately:

(a) 0.2
(b) 1.94
(c) 10
(d) 50

Q8. The solar constant varies over the year by about:

(a) 0%
(b) ±3.3%
(c) ±25%
(d) ±50%

Q9. The sun radiates approximately as a black body at:

(a) 300 K
(b) 1000 K
(c) 5800 K
(d) 15,000 K

Q10. By Wien's law, the peak of the solar spectrum lies near:

(a) 0.10 μm
(b) 0.48 μm
(c) 1.5 μm
(d) 10 μm

Q11. The visible band carries roughly what fraction of solar energy?

(a) 7%
(b) 25%
(c) 47%
(d) 90%

Q12. The ultraviolet portion of the solar spectrum is about:

(a) 7%
(b) 30%
(c) 50%
(d) 70%

Q13. Photosynthetically active radiation (PAR) spans wavelengths:

(a) 0.1–0.2 μm
(b) 0.4–0.7 μm
(c) 1–3 μm
(d) 5–10 μm

Q14. Radiation coming straight from the sun's disc is called:

(a) Diffuse
(b) Beam (direct)
(c) Reflected
(d) Global

Q15. Global radiation on a horizontal surface equals:

(a) Beam only
(b) Diffuse only
(c) Beam + diffuse
(d) Beam - diffuse

Q16. Air mass is defined as:

(a) $\cos \theta_z$
(b) $1/\cos \theta_z$
(c) $\sin \theta_z$
(d) $\tan \theta_z$

Q17. AM0 refers to:

(a) Sea level overhead
(b) Extraterrestrial (outside atmosphere)
(c) 45° sun
(d) Cloudy sky

Q18. The standard reference spectrum for rating PV modules is:

(a) AM0
(b) AM1
(c) AM1.5
(d) AM3

Q19. Atmospheric absorption of solar IR is mainly by:

(a) Nitrogen
(b) Water vapour and CO_2
(c) Argon
(d) Helium

Q20. Ultraviolet radiation is largely absorbed by atmospheric:

(a) Ozone
(b) Oxygen only
(c) Dust
(d) Methane

Q21. Scattering of sunlight by air molecules is:

(a) Mie scattering
(b) Rayleigh scattering
(c) Reflection
(d) Refraction

Q34. PV modules respond to only part of the spectrum because:

(a) Cells absorb all wavelengths

(b) A cell's spectral response is limited (e.g. Si ~0.4–1.1 μm)

(c) There is no IR

(d) UV is 90%

Answers

Q5 — (b) Just outside the atmosphere

Q6 — (b) 1367 W/m² — *WMO value 1367 W/m² (modern satellite value ~1361 W/m²).*

Q7 — (b) 1.94 — *1.94 cal/cm²·min is the classic figure — it corresponds to 1353 W/m²; 1367 W/m² converts to ≈ 1.96.*

Q8 — (b) ±3.3% — *±3.3% over the year — Earth's elliptical orbit, nearest in early January.*

Q9 — (c) 5800 K

Q10 — (b) 0.48 μm — *5800 K → Wien peak ~0.48 μm.*

Q11 — (c) 47% — *Visible ~47% (PAR 0.4–0.7 μm).*

Q12 — (a) 7% — *UV ~7%, visible ~47%, IR ~46%.*

Q13 — (b) 0.4–0.7 μm

Q14 — (b) Beam (direct)

Q15 — (c) Beam + diffuse

Q16 — (b) 1/cos θ_z — *AM = 1/cos θz.*

Q17 — (b) Extraterrestrial (outside atmosphere)

Q18 — (c) AM1.5 — *AM0 space, AM1 overhead at sea level, AM1.5 the PV rating standard (~1000 W/m²).*

Q19 — (b) Water vapour and CO_2

Q20 — (a) Ozone

Q21 — (b) Rayleigh scattering

Q34 — (b) A cell's spectral response is limited (e.g. Si ~0.4–1.1 μm)

Sources: Sukhatme & Nayak — Solar Energy: Principles of Thermal Collection and Storage; Rai — Non-Conventional Energy Sources; Duffie & Beckman — Solar Engineering of Thermal Processes; WMO Guide to Meteorological Instruments (radiation); MNRE/PIB releases (capacity figures confirmed against MNRE data to 30 June 2026); NPTEL renewable-energy lectures.